Network Science

Lecture 05: Social Context and Link Formation

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From Structure to Context

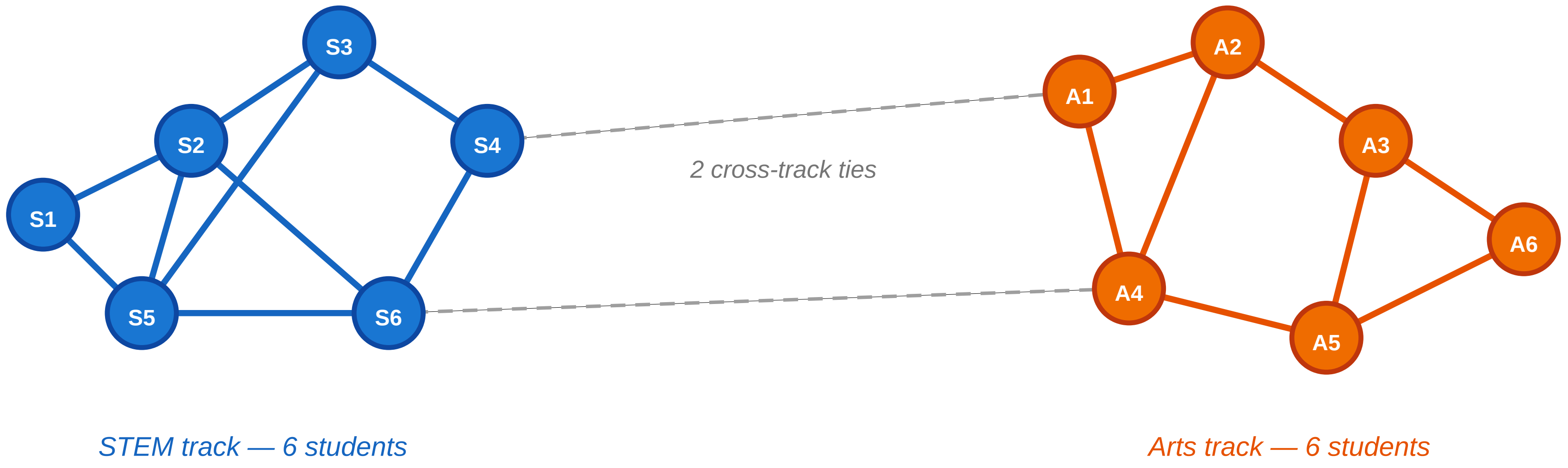
In Lectures 03 and 04 we classified edges (strong vs. weak) and groups (communities). The graph was colorless — every node was interchangeable. Today we ask: what happens when nodes carry *attributes* and edges carry *reasons*?

Consider a small classroom.

A Motivating Scenario

A small classroom: friendships and attributes

circles = students · color = track (blue = STEM, orange = Arts) · lines = friendships



Questions we'd like to answer from this picture

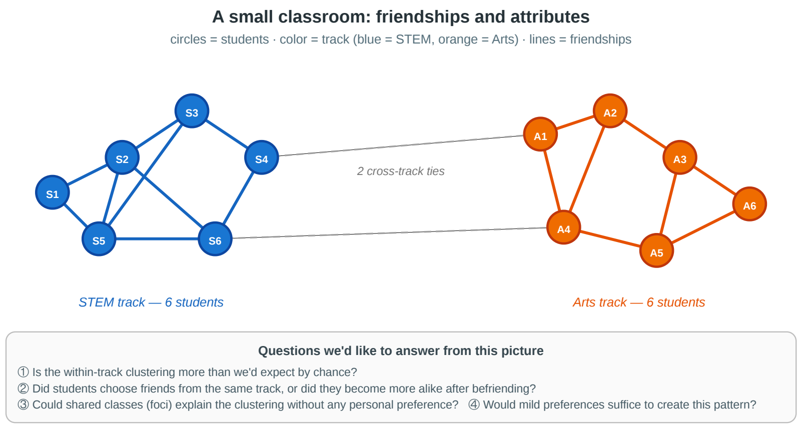
① Is the within-track clustering more than we'd expect by chance?
② Did students choose friends from the same track, or did they become more alike after befriending?
③ Could shared classes (foci) explain the clustering without any personal preference? ④ Would mild preferences suffice to create this pattern?

Twelve students, split equally between STEM (blue) and Arts (orange). Most friendships are within-track; only two cross-track ties exist. The clustering is visible — but why is it there?

Four Questions from One Picture

1. **Measurement.** Is the within-track clustering more than we would expect by chance — or could it happen under random mixing?
2. **Mechanism.** Did students choose friends from the same track (**selection**), or did friends influence each other to pick the same track (**socialization**)?
3. **Context.** Could shared classes or labs (**foci**) explain the clustering without any personal preference for similarity?
4. **Dynamics.** If every student has only a mild preference for same-track friends, could that suffice to produce this sharp segregation?

Every concept in today's lecture answers one of these.



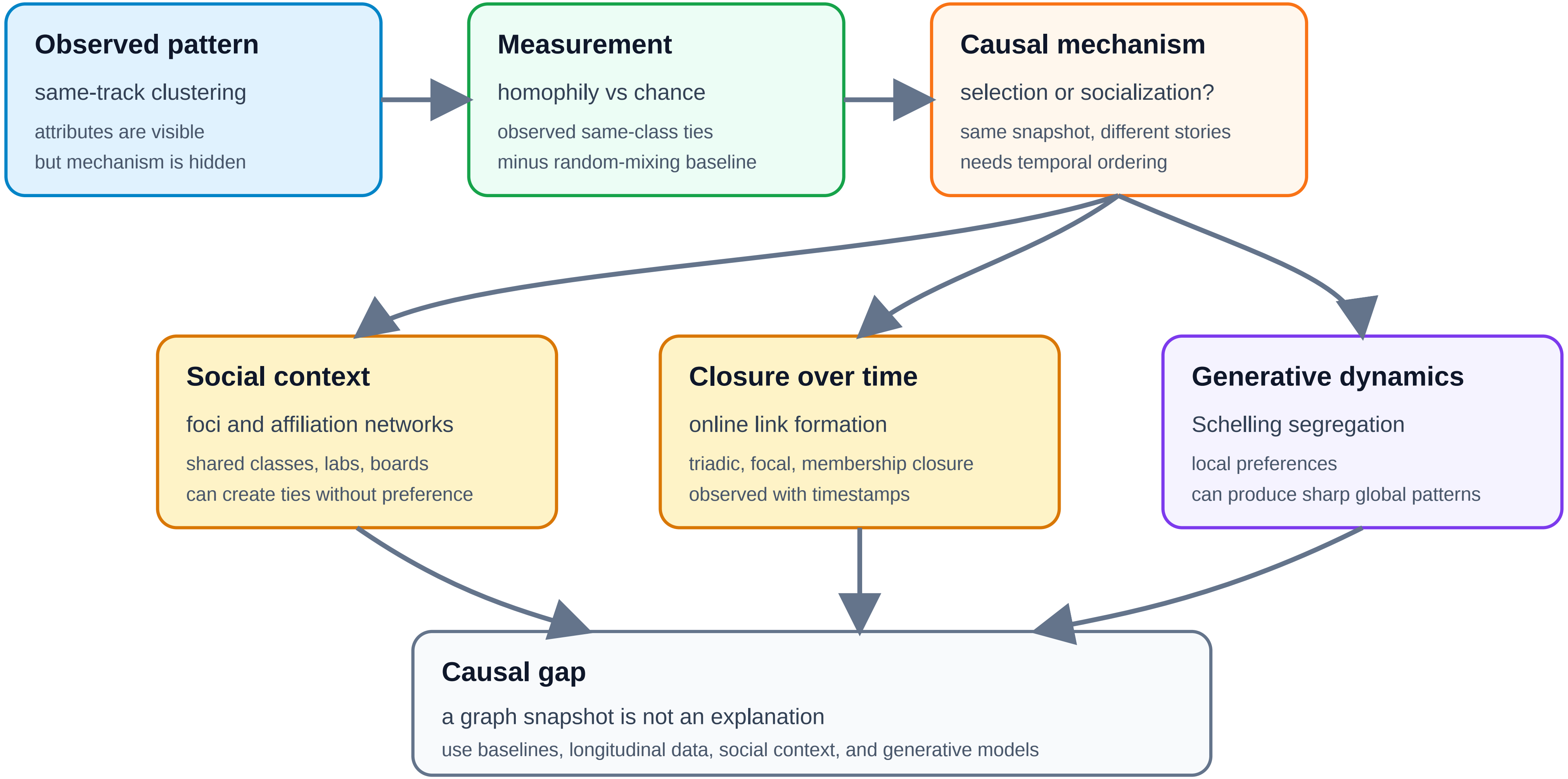
A Different Kind of Gap

In L03–L04 the gap between theory and measurement was **computational**: MaxSTC and modularity maximization are NP-hard, so we used polynomial-time heuristics.

In L05 the gap is **causal**: even with infinite compute, a cross-sectional snapshot cannot tell us *why* a tie formed. The fix is not a better algorithm — it is richer data.

Question	What we want to know	What a snapshot shows	What we need
Homophily?	Is tie formation biased toward similarity?	Within-group edge fraction	Random-mixing baseline
Selection or socialization?	Which came first — similarity or the tie?	Both present simultaneously	Longitudinal data
Foci or preference?	Did shared context cause the tie?	Shared memberships and ties	Affiliation network model
Local → global?	Can mild preferences produce sharp segregation?	A segregated pattern	A generative model (Schelling)

Argument Map



Takeaway

L03–L04 asked "what structure is there?" and hit a computational wall. L05 asks "why is the structure there?" and hits a causal wall. Network analysis requires both algorithmic tools and causal reasoning about the mechanisms behind the observed graph.

Homophily

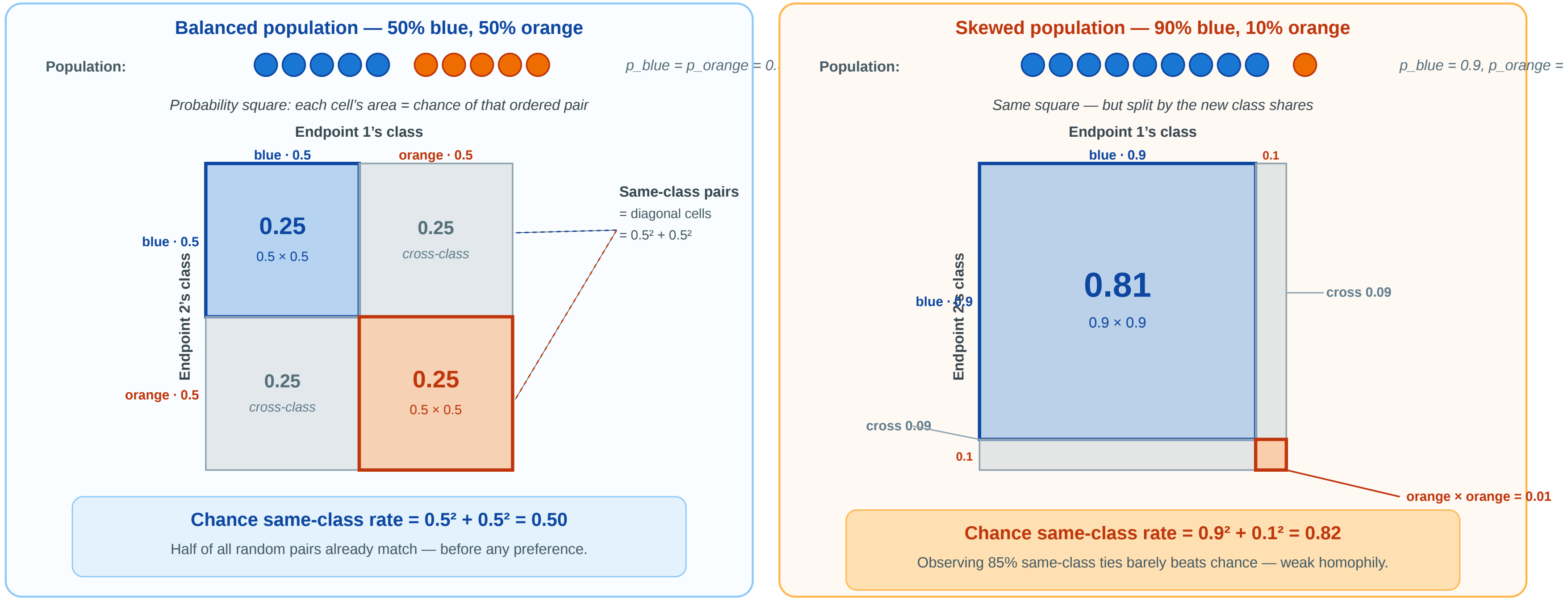
Guiding Question: Do people form ties because they are similar — and how would we know?

What Does Homophily Look Like?

Homophily = tie formation is biased toward similarity. Within-group edges (colored) dominate; cross-group edges (gray) are rare. The pattern looks clear-cut — but a strong within-group share could also arise mechanically when one class is much larger than the other. The next slide makes that worry precise.

Homophily — What Counts as "More Than Chance"?

> Before calling that pattern *homophily*, we have to ask: **how many same-class ties would we see if people paired up purely at random?** Only the *excess* over that chance rate is evidence of a preference for similarity.



Each cell's area is the chance of that ordered pair; the **colored diagonal cells** are the same-class outcomes. In the skewed population (right), 82% of random pairs already match — so an observed 85% within-group rate is almost noise. *Next slide:* we name this chance rate H_{base}.

Homophily — Formal Definition

Homophily. Edge formation is biased toward similarity in a node attribute x. Formally, for a categorical attribute with class shares $p_1, \ldots, p_k$:

- **Random-mixing baseline** (area of the diagonal cells in the probability square):

$$H_{\text{base}} = \sum_{i=1}^{k} p_i^2$$

- **Observed homophily:** the fraction of actual ties that connect same-class nodes, H_{obs}.
- **Homophily index:** how much the observed same-class rate exceeds the random baseline, normalized:

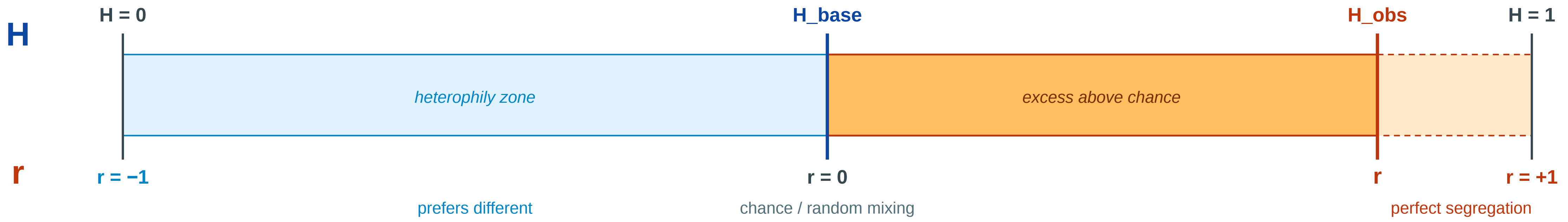
$$r = \frac{H_{\text{obs}} - H_{\text{base}}}{1 - H_{\text{base}}}$$

$r = 0$: random mixing. $r = 1$: perfect segregation. $r < 0$: heterophily (preference for *different* attributes).

Homophily — Three Numbers, One Story

Measure	What it tells you	Depends on
H_{base}	**The chance rate.** What random pairing alone would produce. Large when one class dominates the population.	Class shares only
H_{obs}	**What the data actually shows.** Count the within-class edges and divide by the total.	Observed edges only
r	**How far the network has moved from chance toward perfect segregation**, on a 0–1 scale. $r = 0$: chance. $r = 1$: perfect segregation. $r < 0$: cross-class preference (heterophily).	Both

one bar, two readings: H along the top, r along the bottom



r = (H_obs − H_base) / (1 − H_base)

r < 0 when H_obs falls below the chance baseline H_base

Homophily Baseline Trap

The **same observed within-group share** can mean very different things depending on the population composition. The table below holds $H_{\text{obs}} = 0.85$ fixed and varies the class shares:

Population shares	H_{base}	Room above chance	r	What you can honestly say
50% / 50%	0.50	0.50	0.70	Strong excess — 70% of the way to perfect segregation
70% / 30%	0.58	0.42	0.64	Substantial — 64% of available room used
90% / 10%	0.82	0.18	0.17	Barely above chance — almost nothing to explain
99% / 1%	0.98	0.02	< 0	*Below* chance — actually heterophilic!

Literature example. McPherson, Smith-Lovin & Cook distinguish **baseline homophily** from **inbreeding homophily** (McPherson et al. 2001). A school can show many same-age friendships simply because classes, lunch periods, and activities are age-graded; a workplace can show many same-occupation ties because people in the same job category share meetings and tasks. The real homophily question is not "are many ties same-type?", but "are there **more** same-type ties than the opportunity structure already predicts?"

> Never interpret "X% of ties are within-group" without asking: **X% compared to what baseline?** The baseline depends on the population, not on what feels intuitively high.

Back to the Classroom

Both tracks have 6 students, so $p_{\text{STEM}} = p_{\text{Arts}} = 0.5$.

- $H_{\text{base}} = 0.5^2 + 0.5^2 = 0.50$
- 17 within-track edges of 19 total: $H_{\text{obs}} = 17/19 \approx 0.89$
- $r = (0.89 - 0.50)/(1 - 0.50) = 0.78$

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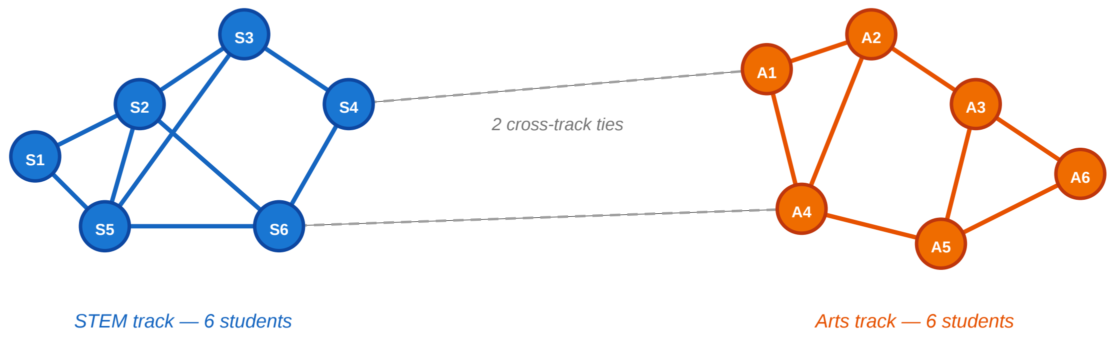

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$r = 0.78$ is strong — far above random mixing. But it does not tell us *why*. Is it selection, socialization, or shared context?

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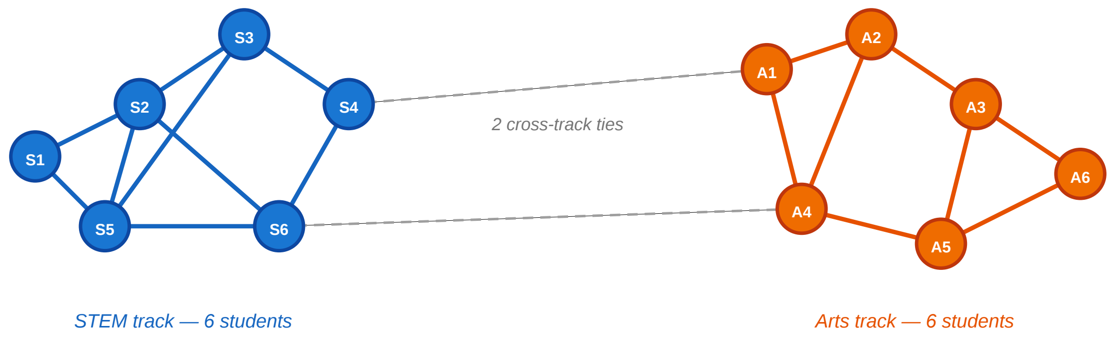

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Quick Check

In a school of 200 students, 60% are in track A and 40% in track B. Of all friendships, 78% are within-track.

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3. Is r strong evidence of homophily? What could make it misleading?

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3. $r = 0.54$ is moderate-to-strong evidence of homophily. It could be misleading if:

- the two tracks occupy different buildings (shared context, not personal preference);
- students were *assigned* to tracks based on pre-existing friend groups (reverse causation);
- a third factor (e.g. family background) causes both track choice and friendship patterns (confounding).

Echo Chambers — The Hypothesis

Source: Cinelli et al. 2021

Claim	Online platforms expose users mostly to like-minded peers and sources.
Network signal	High homophily in the *interaction* network (likes, retweets, replies, comments) when nodes are coloured by political leaning.
Diffusion signal	Information circulates *within* groups rather than crossing them.

Why it's a homophily story. An echo chamber is not just "people who disagree online" — it is a region of the interaction graph where within-group ties dominate so heavily that the same content recirculates without leaving the cluster.

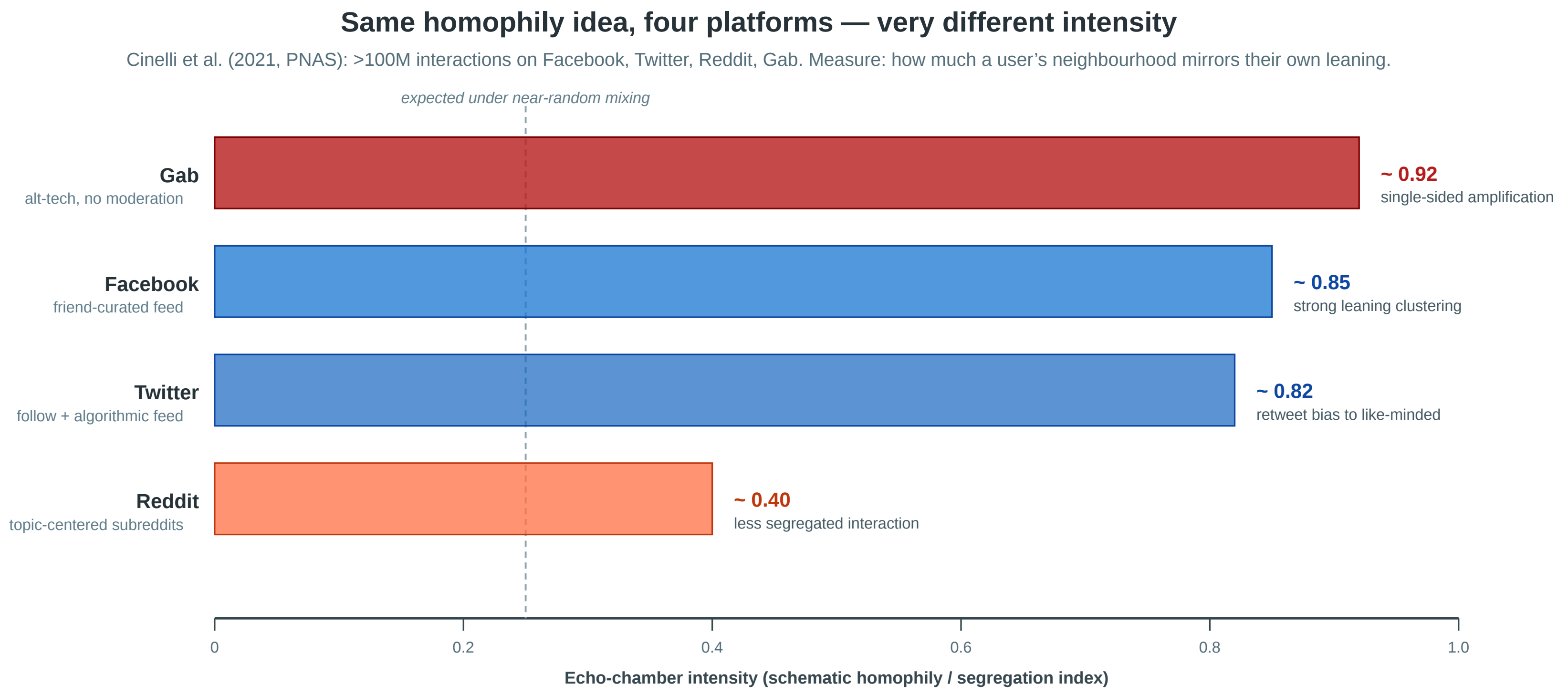
The empirical test (Cinelli et al. 2021, PNAS).

- Four platforms: **Facebook, Twitter, Reddit, Gab**.
- Over **100 million** interactions, hundreds of news outlets coded by political leaning.
- For each user i, infer a leaning score x_i as the **average leaning of the scored posts, links, pages, or sources** they produce or endorse; then compute the average leaning of their interaction neighborhood: $x_i^N = \frac{1}{k_i^{out}} \sum_j A_{ij} x_j$.
- If $x_i \approx x_i^N$, the user mostly interacts with like-minded peers.

So x_i is a behavioral estimate: it summarizes the political leaning of the content a user chooses to post, like, share, or link to.

Different platform architectures (algorithmic feed, friend graph, subreddit topic, alt-tech) let us see whether echo chambers are universal or **architecture-dependent**.

Echo Chambers — Same Idea, Four Platforms



Schematic. Magnitudes are illustrative; the paper reports leaning-gap distributions, not a single index. The qualitative ranking (Gab > Facebook ≈ Twitter ≫ Reddit) matches the paper.

Architecture matters. Facebook and Twitter (friend / follow graphs amplified by ranking) show strong homophilic clustering by political leaning. **Gab** — small, unmoderated, single-sided audience — is even more extreme. **Reddit** is much less segregated: topic-centered subreddits force users to interact across

Takeaway

Homophily is not just a visual pattern — it is a statistical claim that tie formation is biased toward similarity *relative to a random-mixing baseline*. The homophily index r quantifies the excess, but it cannot identify the mechanism.

Selection and Socialization

Guiding Question: How can we distinguish whether similarity caused the tie or the tie caused the similarity?

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$$\text{homophily} = \text{selection} + \text{socialization} + \text{context}$$

Three Mechanisms, One Pattern

Selection. People choose to form ties with others who are *already* similar to them. Similarity precedes the tie.

Socialization (social influence). People become *more similar* to their contacts after the tie forms. The tie precedes similarity.

Contextual correlation. A shared environment (school, workplace, neighborhood) independently causes both the attribute and the tie. Neither causes the other directly.

Same Snapshot, Three Stories

Suppose we observe: **STEM students mostly befriend STEM students.**

Mechanism	Concrete story	What data would help?
Selection	Students first choose STEM, then choose friends from the same track.	Friendships observed *after* track choice
Socialization	Friends influence each other to choose the same elective or track.	Track changes observed *after* friendships
Contextual correlation	STEM students share labs, schedules, and buildings, so they meet more often.	Timetables, room assignments, shared courses

> The graph snapshot is identical in all three stories. The difference is the **time order** and the **unobserved context**.

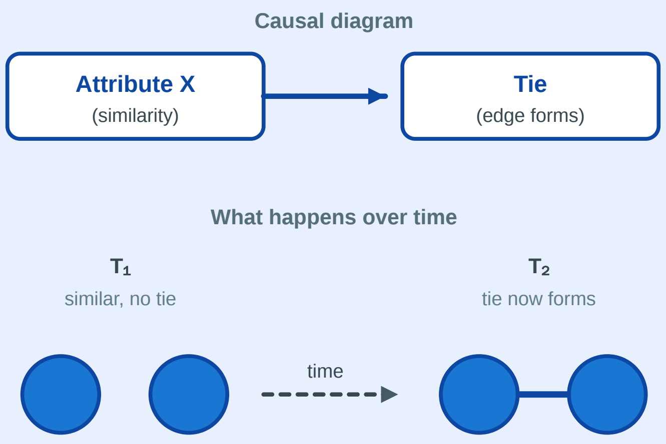
Causal Diagrams

Three causal mechanisms — same observed pattern

All three produce the cross-sectional fact “connected nodes are similar.” Distinguishing them needs temporal data or knowledge of hidden context.

A. Selection

similarity → tie



Classroom example

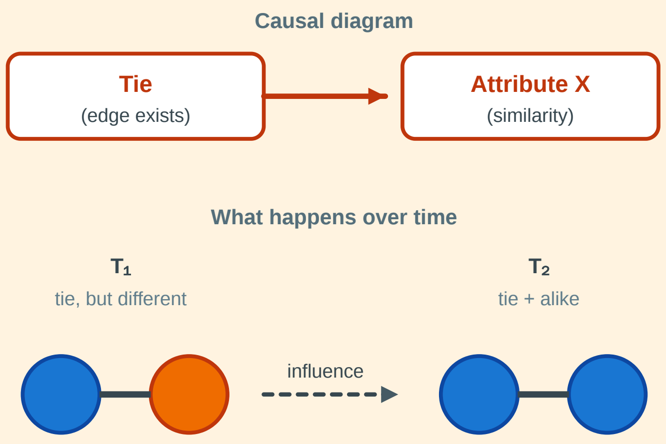
Two students are already in STEM. They meet and become friends *because* they share that interest.

Empirical signature:

new ties form preferentially between already-similar people.

B. Socialization (influence)

tie → similarity



Classroom example

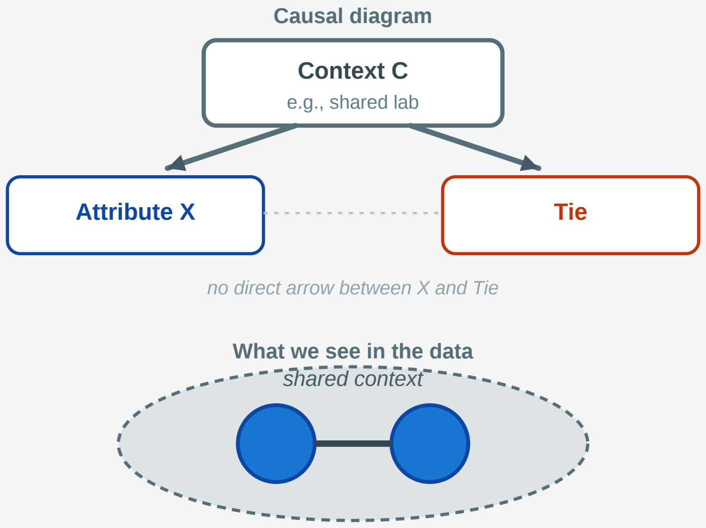
A STEM and an Arts student become friends. Over months, the Arts student shifts interests toward STEM.

Empirical signature:

attributes change after ties form, in the direction of friend's attribute.

C. Contextual correlation

C → similarity AND C → tie



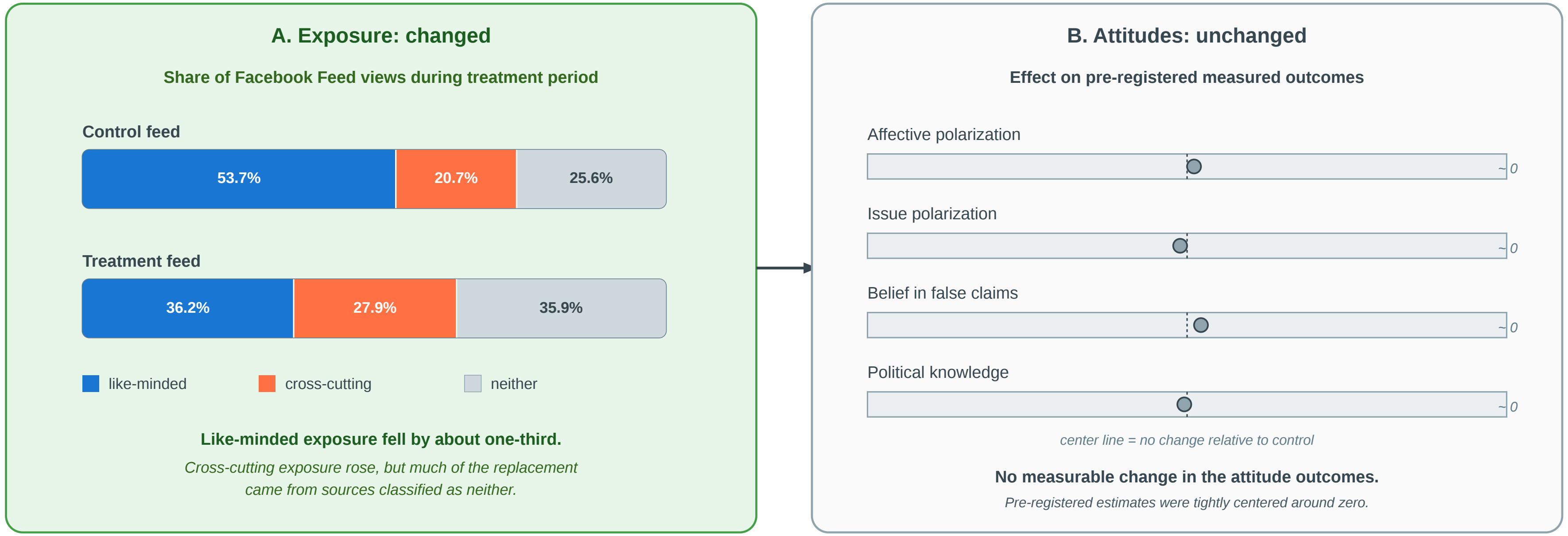
Classroom example

Two students share a lab. The lab independently makes them similar **and** brings them together.

Empirical signature:

association vanishes when conditioning on C.

Empirical Test: Does Exposure Cause Polarization?



Selection question: do already-polarized users choose like-minded friends, pages, and platforms?
Socialization question: does repeated exposure to like-minded content make users more polarized?

Source: Nyhan et al. 2023

Nyhan et al. (2023, Nature) isolate the second pathway. In a pre-registered Facebook field experiment during the 2020 U.S. election, **23,377 consenting users** were randomly assigned to a feed intervention that reduced politically like-minded source exposure by about one-third. The treatment increased cross-cutting exposure, but had **no measurable effect** on eight pre-registered attitude outcomes, including affective polarization, ideological extremity, candidate evaluations, and belief in false claims.

> **Interpretation.** The experiment weakens one short-run **socialization** story: changing feed exposure for three months did not measurably change attitudes. It does **not** test **selection** into like-minded networks or platforms, because users' existing friends, follows, and platform choices were not randomly assigned.

Empirical Test: Christakis & Fowler — Obesity in Social Networks

Christakis & Fowler (2007), *The Spread of Obesity in a Large Social Network Over 32 Years*, New England Journal of Medicine 357, 370–379.

They analyzed the **Framingham Heart Study** social network — 12 067 people tracked over **32 years** with repeated measurements of both friendship ties and body mass index. This made it possible to observe the *temporal sequence* of tie formation and weight change.

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Headline result. A person's probability of becoming obese increased by **57%** if a friend became obese in the same interval. The effect was stronger for mutual friends and showed **directionality** — it operated in the direction of the friendship nomination, suggesting socialization (influence) rather than pure selection.

What the Study Could Not See

1. Confounders. Cohen-Cole & Fletcher (2008) and Lyons (2011) showed that the same statistical method would "detect" social contagion of **height** and **acne** — traits that clearly do not spread through social influence. The method may confuse *shared environment* with *social influence* when the confounders evolve over time.

2. Dynamic selection. The study controlled for whether two friends were already similar in weight. But people may also **choose new friends**, or keep reporting old friends, because their lives and weight trajectories are becoming similar. Then "my friend became obese before me" may partly reflect changing friendship selection, not only social influence.

3. Missing network. The Framingham data records nominated friends, not the full social network. Influence may flow through unobserved ties (coworkers, family, neighbors) that correlate with both friendship nominations and weight change.

Quick Check

You observe that smokers tend to befriend other smokers in a college dorm.

1. What kind of data would you need beyond a single snapshot to distinguish selection from socialization?
2. Sketch one observation pattern that would favor *selection* and one that would favor *socialization*.
3. Even with longitudinal data, what kind of confound would remain hard to rule out?

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1. Longitudinal observations — friendships and smoking status at multiple time points, with newly formed ties and newly adopted behavior visible.

2.

- *Selection:* new friendships disproportionately form between people who **already** share smoking status.
- *Socialization:* people who befriend smokers become more likely to **start** smoking in the next observation period.

3. Shared *environment* — e.g. dorm floor placement causing both friendship and smoking exposure simultaneously. Even with temporal data, confounders that evolve over time alongside

Takeaway

Cross-sectional data cannot distinguish selection from socialization — the same homophily index r is consistent with both. Longitudinal network data is far more informative, but even multi-decade panel studies cannot fully eliminate confounding by evolving shared environments.

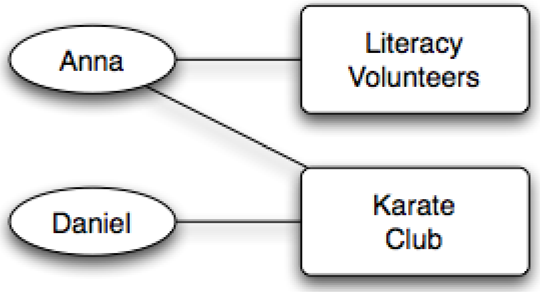
Affiliation Networks

Guiding Question: How do we represent the social settings — classes, clubs, workplaces — in which ties form?

Bipartite Affiliation Model

An **affiliation network** is a bipartite graph $G = (P \cup F, E)$:

- P: actor nodes (persons)
- F: focus nodes (groups, locations, classes, organizations)
- E: edges only between P and F — "person p participates in focus f"



Source: Easley & Kleinberg 2010

The focus nodes are the **social context**: they encode where people meet, what they jointly attend, and which settings create repeated exposure. This lets us separate "people are similar" from "people share the same places."

Shared foci do not prove a tie. They create opportunities for contact, information, and influence.

From Affiliations to Co-Occurrence Graphs

The bipartite graph is the source of truth. To use standard network tools, we often project it into a one-mode **co-occurrence graph**.

Let B be the person-by-focus membership matrix:

$$B_{pf} = 1 \quad \text{if person } p \text{ participates in focus } f.$$

Then:

- **Person co-occurrence:** $BB^\top$
- **Focus co-occurrence:** $B^\top B$

An edge weight counts **how many contexts are shared**.

Projection	Nodes	Edge weight means	Then analyze with
$BB^\top$	People	shared classes, clubs, boards, workplaces	centrality, communities, clustering
$B^\top B$	Foci	overlapping participants	similarity, clusters, diffusion channels

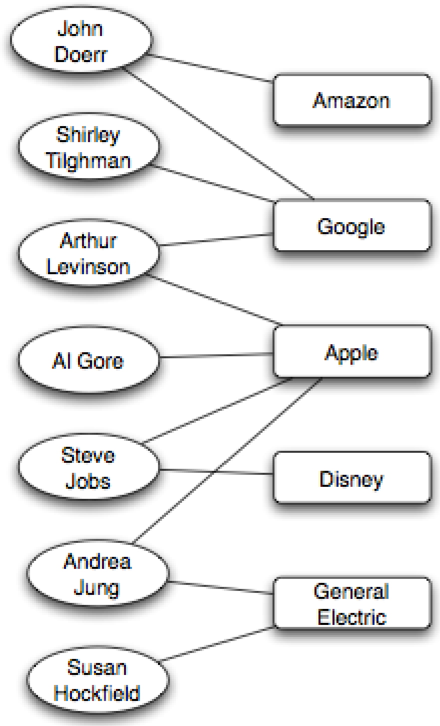
> Co-occurrence is **context**, not a confirmed relationship. It says two units had an opportunity to interact.

Reading the Board Network

Corporate boards are an affiliation network: board members connect to company boards. The co-occurrence projection decides what counts as a node and what the shared context means.

Co-occurrence graph	Edge means	Use it for
Board-member graph	Two board members share ≥ 1 company board	Peer exposure, governance norms
Company graph	Two firms share ≥ 1 board member	Strategic interlocks, diffusion across firms

Example — Corporate Board Interlocks

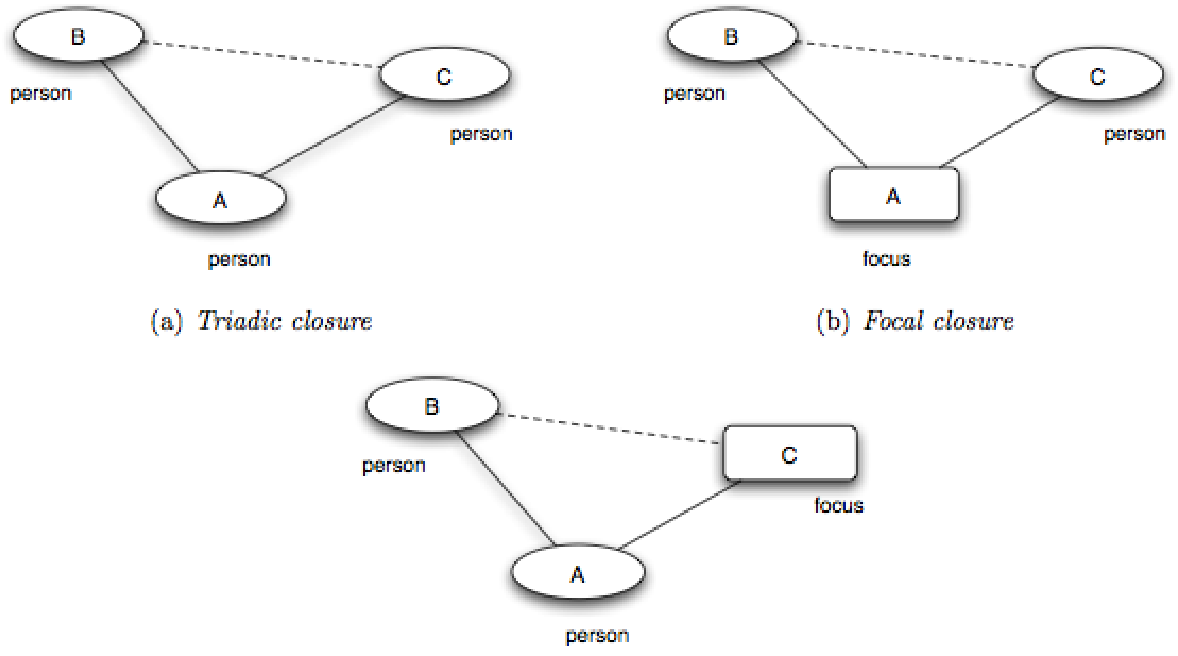


Source: Easley & Kleinberg 2010

Reading rules.

1. A projected edge means **shared context**, not friendship or influence.
2. Projection creates dense cliques: a board with s members adds $\binom{s}{2}$ member-member edges.
3. Keep weights: sharing five boards is stronger evidence than sharing one.

Three Closure Processes



Source: Easley & Kleinberg 2010

Triadic closure. Friend-of-friend becomes friend — a person–person–person path closes into a triangle. *(Same mechanism as Lecture 03.)*

Focal closure. Two persons sharing a focus become friends — a person–focus–person path closes into a person–person tie. *(New — the focus mediates the introduction.)*

Membership closure. A friend introduces you to a focus you did not belong to — a person–person–focus path closes into a person–focus tie. *(New — the friend mediates membership.)*

Closure Processes — Classroom Examples

Closure type	Open path	What closes?	Classroom example
Triadic	student → friend → student	A new friendship	"My lab partner introduces me to their friend."
Focal	student → elective → student	A new friendship	"We sit in the same statistics elective and start talking."
Membership	student → friend → club	A new membership	"My friend invites me to join the robotics club."

Only triadic closure needs a shared friend. Focal closure can create ties among strangers through repeated exposure in the same setting.

Quick Check

For each scenario, identify whether triadic, focal, or membership closure is at work.

1. Two students take the same elective and become friends.
2. A new graduate student joins the same lab as her advisor's other student, who introduces her.
3. A friend invites you to their book club, and you join.

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1. Focal closure — the elective is the shared focus that brings them into contact.

2. Triadic closure — the advisor is the shared friend that closes the triangle.

3. Membership closure — your friend brings you into a focus you did not previously belong to.

Takeaway

Affiliation networks make social context explicit. Focal closure and membership closure are distinct tie-formation mechanisms that can produce homophily-like patterns without any personal preference for similarity — making them a critical alternative explanation in any homophily analysis.

Online Link Formation

Guiding Question: What can large-scale online data teach us about how ties form over time?

The Setting — Kossinets & Watts (2006)

Data. A large U.S. research university's internal email logs, ca. 1 year, **~43,000 students/faculty/staff**, **~14 million** messages — plus full course-enrollment records and organizational-affiliation rosters.

Why it matters. The timestamps make the analysis forward-looking: we can observe what existed at time t, then ask which new email ties form at $t+1$.

Used here for two experiments.

- **Triadic closure:** shared email contacts.
- **Focal closure:** shared classes / institutional foci.

The membership-closure plots that follow are separate Easley-Kleinberg examples from LiveJournal and Wikipedia, not the same Kossinets-Watts university dataset.

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> The timestamped data separates *"what existed at t"* from *"what formed by $t+1$"* — the missing ingredient in homophily snapshots.

Revisit Triadic Closure first: Independent-Opportunity Baseline

Before reading the empirical curve, we should think of a baseline. What could that be?

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$$P(\text{new tie} \mid k \text{ shared friends}) = 1 - (1-q)^k$$

Shared friends k	1	2	3	5
P if $q = 0.20$	0.20	0.36	0.49	0.67

Why this form?

- q: probability, one mutual friend creates an opportunity.
- $1-q$: that mutual friend does **not** create one.
- $(1-q)^k$: none of the k mutual friends creates one.
- $1-(1-q)^k$: at least one opportunity appears.

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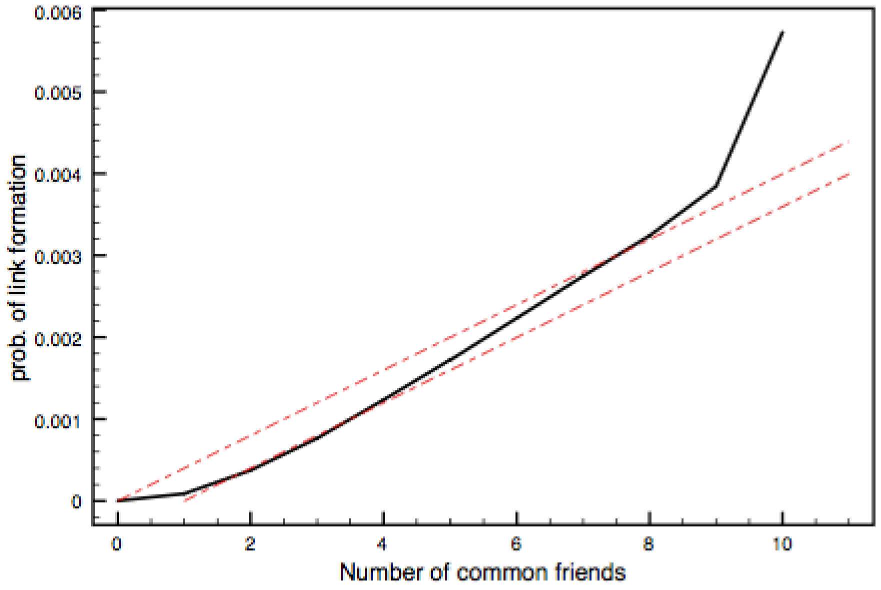
What does this mean? The model gives us a **comparison curve**, not a claim that friends really act independently. The table uses a large q only to make the curvature visible. In the Easley-Kleinberg figure, q is tiny, so the curve looks almost linear on the plotted range.

In the empirical figure, the second dotted curve is the same baseline shifted one step right: it asks what the curve would look like if the first mutual friend had a smaller effect.

Triadic Closure at Scale

Black curve: observed probability $T(k)$. Red dotted curves: independent-opportunity baselines from the previous slide. (Upper/Lower: $1-(1-q)^k$; $1-(1-q)^{k-1}$; $q \approx 4 \times 10^{-4}$)
Because q is very small here, $1-(1-q)^k \approx kq$ over the plotted range, so the dotted baseline appears nearly linear even though the exact function is concave.



Source: Easley & Kleinberg 2010

What was measured. At each time step, take every pair (u, v) that has **not yet emailed**.
Count k: how many mutual email contacts they already share. Then ask whether (u, v) exchanges email in the next interval.
The output is a curve: $P(\text{new tie})$ vs. k.

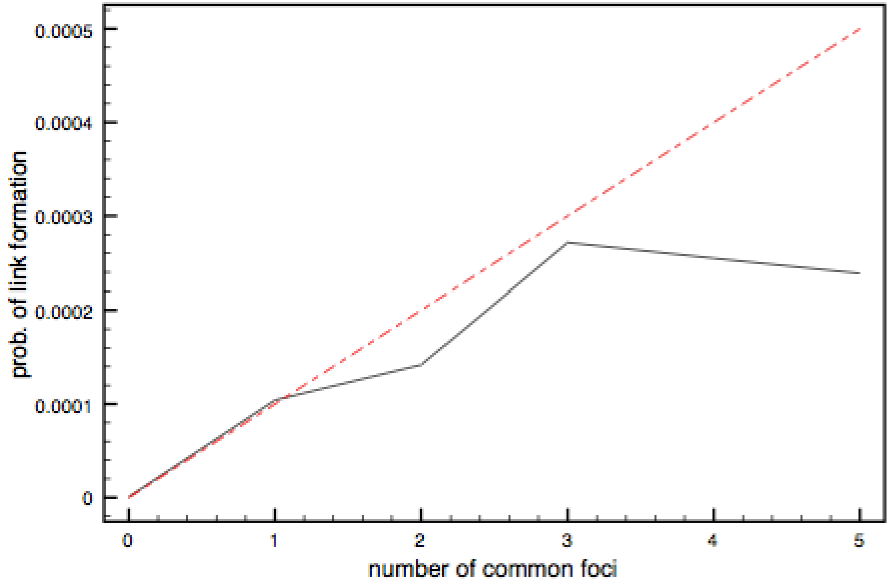
k shared friends	$P(\text{new tie})$
0	very low (baseline)
1	sharply higher
2	higher still
5--8	roughly follows the independent-friend baseline
9+	rises steeply again, but with fewer observations

1. **Mechanism.** Mutual friends create *opportunities*: introductions, shared interests, social pressure to know each other.
2. **Shape.** The curve is not just "more friends, linearly more probability." It is close to the independent baseline in the middle, but bends upward at high k; those high-k points should be read cautiously because there are fewer pairs with many mutual friends.

Do not read the red-dotted lines as confidence intervals. They are toy-model baselines, not two fitted values of q. The value of q only puts the comparison curve on the observed scale; it is not a causal estimate of one mutual friend's effect. Also be cautious at high k: pairs with many mutual friends are rare, so the curve can be noisier there.

Focal Closure at Scale

Black curve: observed probability. Red dotted curve: independent-focus baseline, $T_{\text{base}}(k) = 1 - (1-p)^k$, with p calibrated from the effect of one shared class.



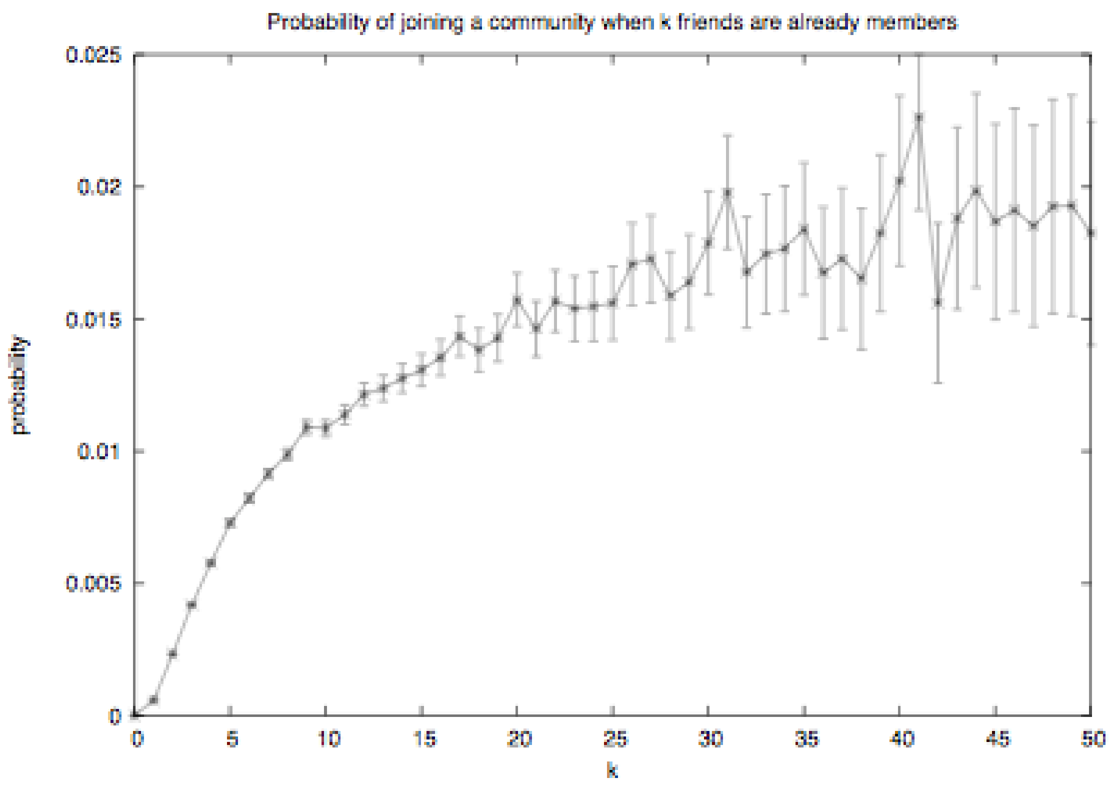
Source: Easley & Kleinberg 2010

Setup. Two people who have **never emailed each other** but co-enroll in k classes during the same semester. What is the probability they exchange email within the term?
Result. One shared class is already informative. But after that, the observed black curve stays **below** the independent-focus baseline: additional shared classes add less than the toy model predicts.

- Focal closure runs **without any shared friend**. Strangers in the same context form ties through repeated exposure alone.
- This is the mechanism that turns "people share classes" into "people share friendships" — and it produces homophily-like patterns even when there is **no preference for similarity at all**.
- It is L05's main answer to *Question 3* from the opening slide ("could shared context explain the clustering?"). Often, yes.

Interpret the dotted line as a baseline, not a fitted causal law. The gap below it means shared classes are partly redundant: taking five classes together is not five independent chances to become connected.

Membership Closure at Scale



Source: Easley & Kleinberg 2010

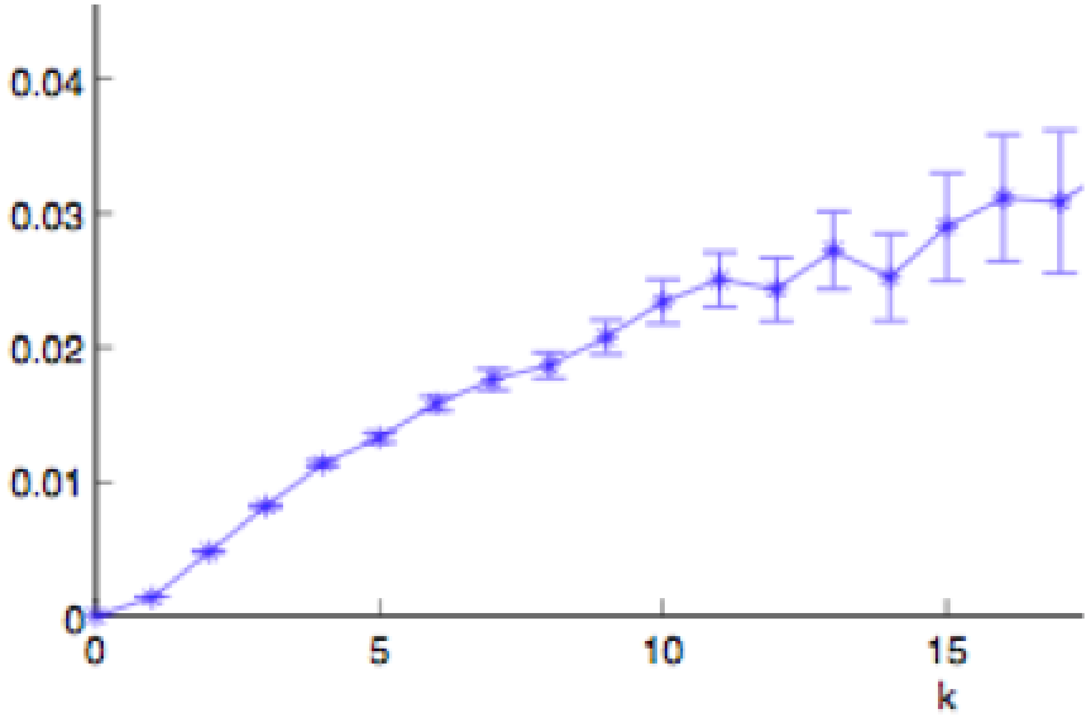
Setup. A person is **not yet a member** of a focus (course, club, group). How does the probability of joining depend on the number of existing members who are already their friends?

Result. Same saturating shape. One friend already in the focus is a strong predictor of joining; additional friends compound — but with diminishing returns.

The flipped direction.

- Triadic and focal closure close *person–person* ties.
- Membership closure closes a *person–focus* tie — your friends pull you **into** the contexts they already inhabit.
- This is how **social networks shape affiliations**, not just the other way around. Friendships and foci co-evolve through both directions of closure.

Wikipedia Editing as Membership Closure



Source: Easley & Kleinberg 2010

Dataset. Wikipedia edit histories plus editor communication. Editors are people; articles are foci; user-talk interactions define editor-editor social ties.

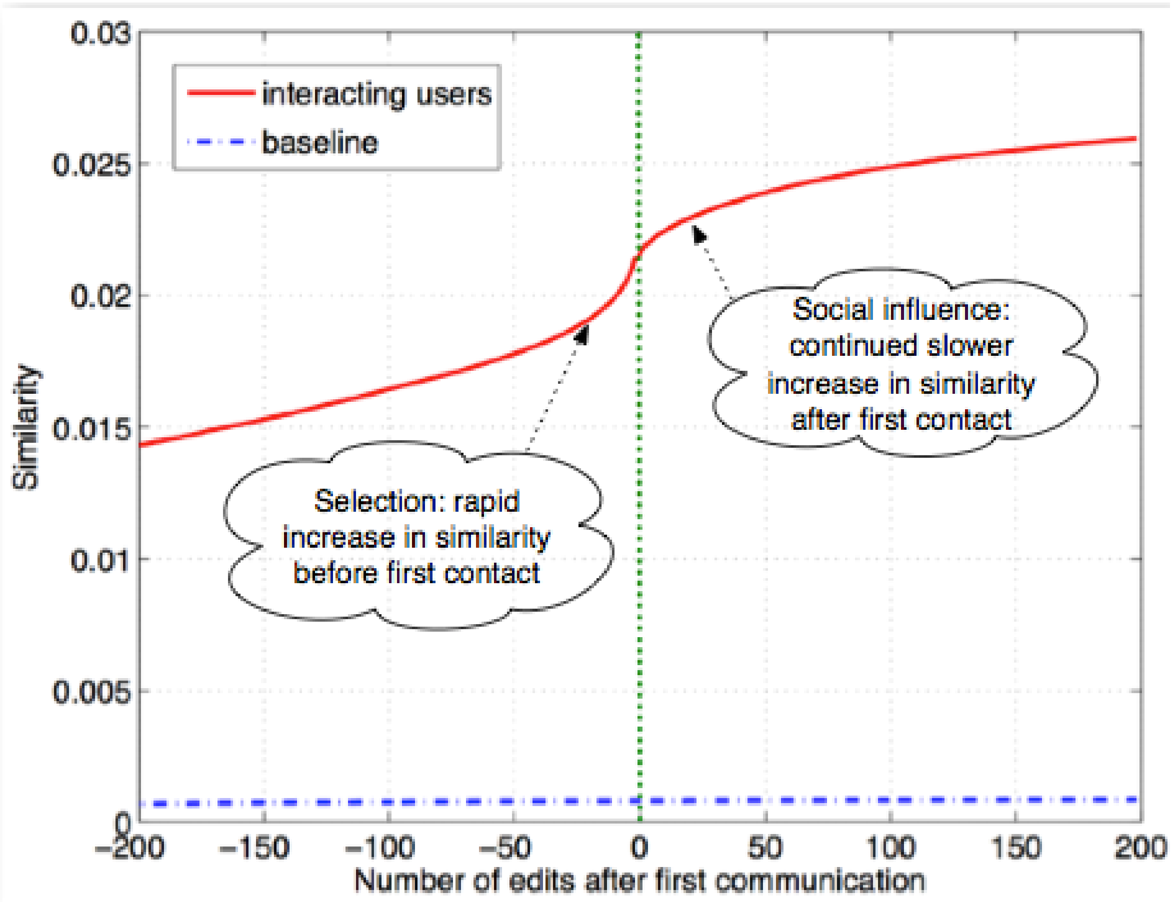
Setup. An editor is connected to an article after editing it.

For an editor who has **not yet edited** an article, count k: how many of their communication partners have already edited that article?

Result. The probability of editing the article rises with k. Friends already active in a focus make joining that focus more likely, with diminishing marginal returns.

> This is membership closure with a shared knowledge object as the focus: friends already active around an article make later editing of that article more likely.

Wikipedia Editors: Selection and Socialization



Source: Easley & Kleinberg 2010

Dataset. Wikipedia edit histories plus first user-talk communication between pairs of editors (Crandall et al. 2008).

Design. Align editor pairs at time 0: the moment they first communicate. Then track similarity in edited articles before and after that first contact.

Reading.

- Before first contact, future communication partners already become more similar than baseline: **selection**.
- After first contact, similarity continues rising more slowly: possible **socialization**.
- This is a feedback loop: similarity makes interaction more likely, and interaction can further increase similarity.

> The figure separates timing, not perfect causality. It strongly suggests both selection and socialization, but unobserved shared contexts can still contribute to the pattern.

Quick Check

A study finds that the probability of two coworkers becoming Facebook friends rises from 1% (no shared friends) to 20% (one shared friend) to 30% (two) to 33% (three or more).

1. Describe the shape of the curve.
2. Fit the independent-opportunity model: what value of q gives $P = 0.20$ at $k = 1$?
3. Does the model predict the observed $P = 0.30$ at $k = 2$? What does the mismatch tell you?

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1. Steep initial rise from 1% to 20%, then much smaller gains (30%, 33%) — a **saturating** curve.

2. At $k = 1$: $1 - (1 - q) = q$, so $q \approx 0.20$.

3. The model predicts $1 - (1 - 0.20)^2 = 0.36$ but we observe **0.30** — a shortfall. Shared friends are *correlated* (typically embedded in the same group), so they provide less independent opportunity than the model assumes. Diminishing returns are *steeper* than independence predicts.

Takeaway

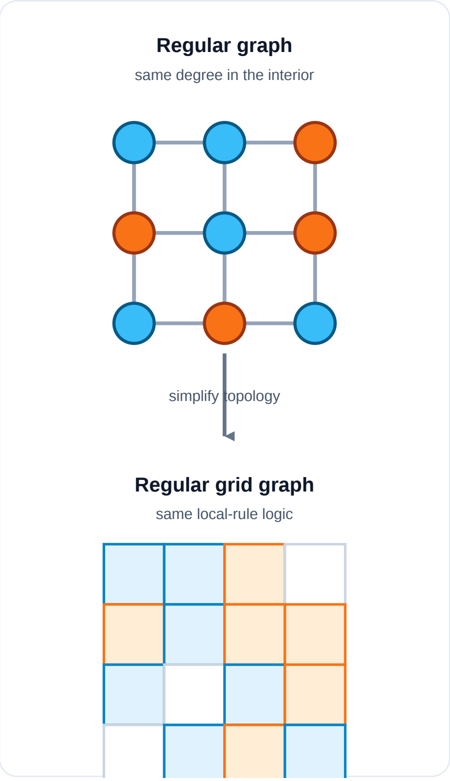
Online data turns link formation into a measurable, time-stamped process. Empirical closure curves show saturation — the first shared contact matters most, and additional contacts are redundant rather than independent.

From Local Preferences to Global Segregation

Guiding Question: Can weak local preferences generate strong global segregation?

Segregation as a Graph

Segregation means that groups are separated across neighborhoods or network regions: most local contacts are same-type, and cross-type contact is rare. It is a **global pattern** in the whole graph, not just one person's preference.



Schelling turned homophily into a dynamic model: **how can small local preferences for similar neighbors accumulate into large-scale segregation?** The model is useful because it separates a local rule from the global pattern it produces.

Model: The graph version makes the mechanism explicit:

- **nodes** are agents or households
- **edges** are neighborhood/contact relations
- a **grid** is a regular graph: each interior cell has the same local neighborhood size
- **node type** is the attribute that matters locally
- **updating** means moving on a grid or rewiring ties in a network

Schelling's Threshold Model

Setup. Agents of two types are arranged on a grid (or network). Each agent i has a threshold $\tau \in [0, 1]$ — the minimum fraction of same-type neighbors it needs to be "satisfied".

Update rule.

1. Identify all *unsatisfied* agents (fewer than fraction τ of neighbors are same-type).
2. Each unsatisfied agent moves to a random vacant cell (or rewires a random tie in a network setting).
3. Repeat until no agent wants to move — or until a stable oscillation is reached.

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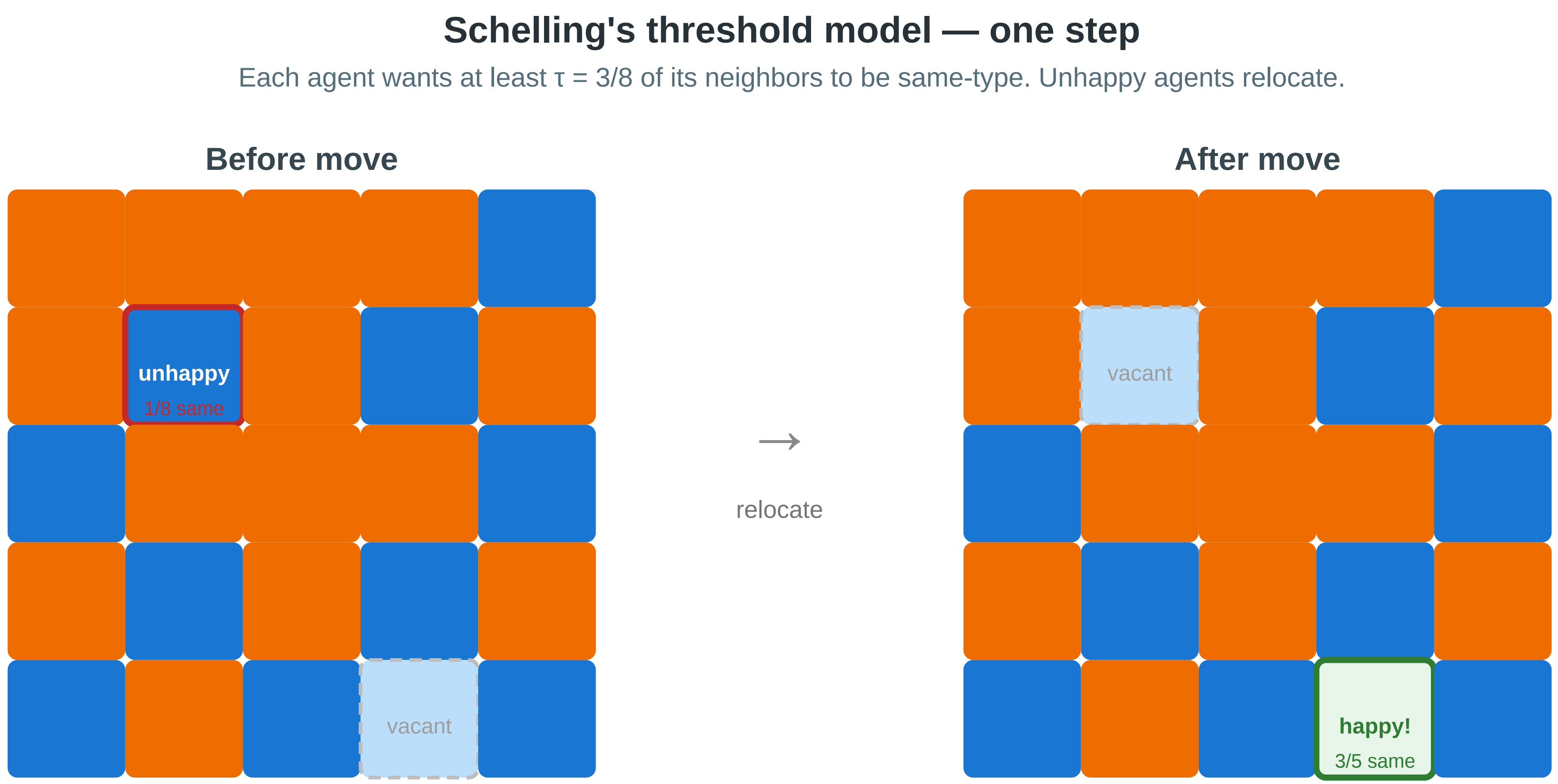
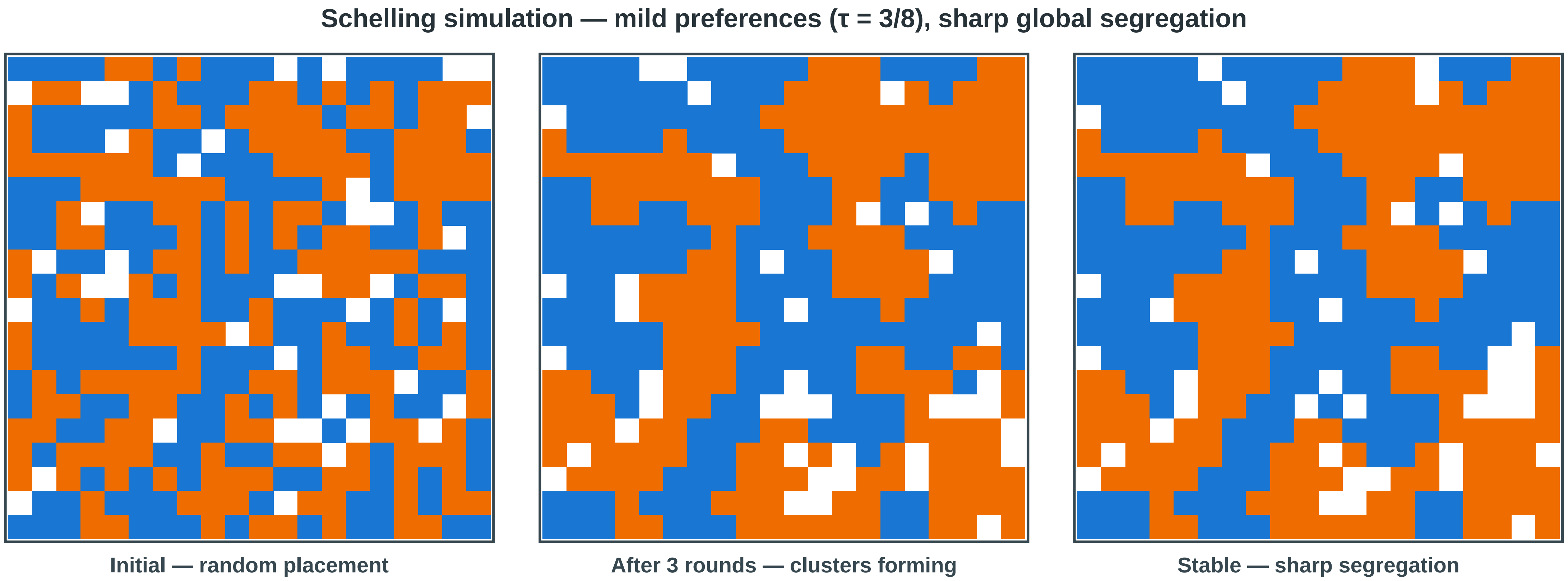
Schelling's insight (1971). Even τ as low as $1/3$ — "I just want a third of my neighbors to be like me" — produces **sharply segregated** global patterns.

One Step of the Model



The blue agent at (1,1) has 1/8 same-type neighbors — below $\tau = 3/8$. It relocates to vacant cell (4,3), where 3/5 neighbors are same-type — satisfied. Repeat across all unsatisfied agents until equilibrium.

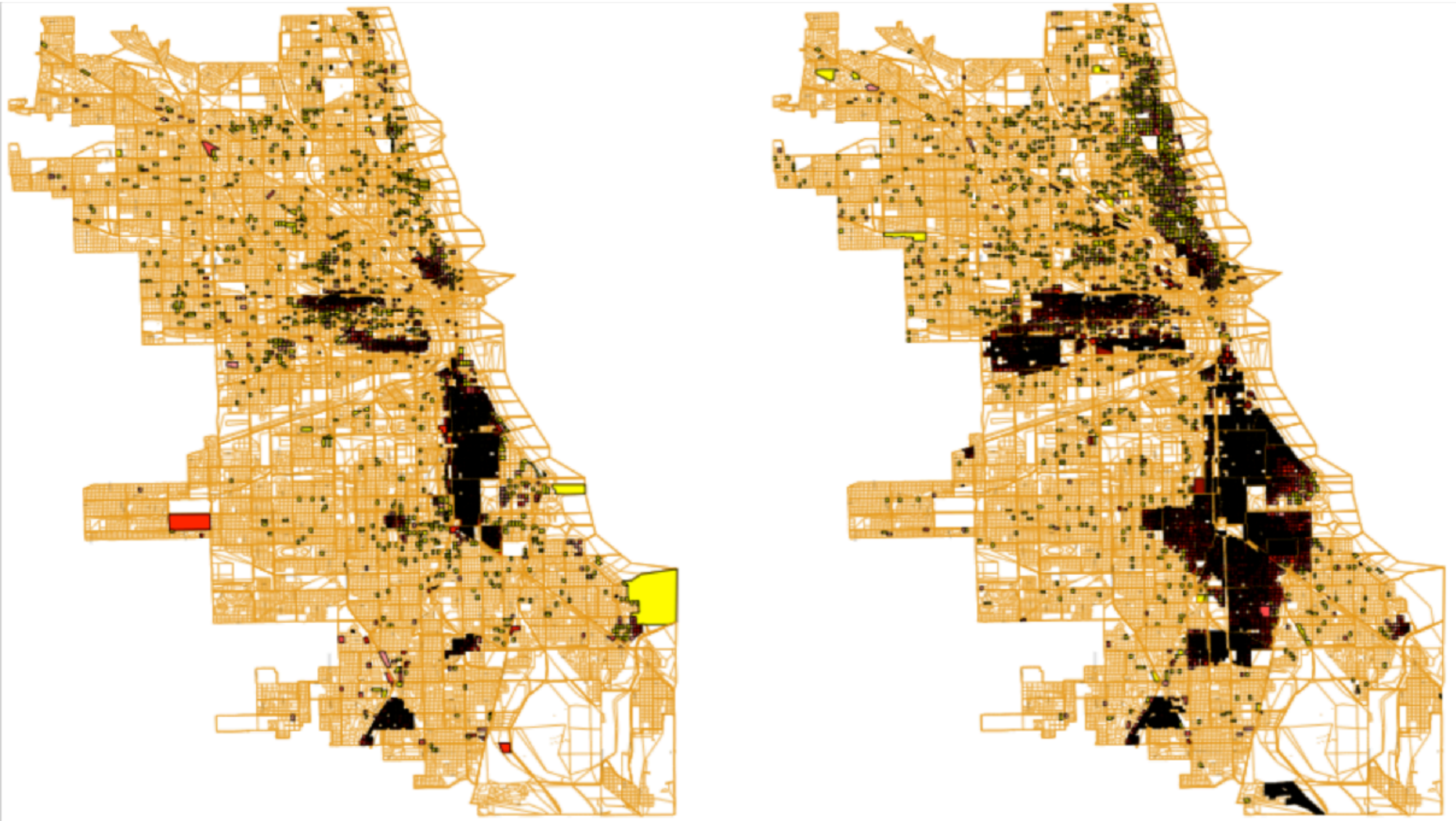
Schelling Simulation



Source: Own Schelling simulation, in the spirit of Easley & Kleinberg (2010), Fig. 4.19

A 20×20 grid with $\approx\!45$ blue, $\approx\!45$ orange, $\approx\!10$ vacant, run with threshold $\tau = 3/8$. Each round, every unsatisfied agent moves to a vacant cell where it *would* be satisfied; the process halts when nobody wants to move. Within a handful of rounds, the locally-mild rule produces **macroscopically sharp** boundaries — exactly the picture the empirical map shows next.

Empirical Motivation



(a) *Chicago, 1940* (b) *Chicago, 1960*

Source: Easley & Kleinberg 2010

Residential segregation in Chicago by race/ethnicity. The sharp boundaries between neighborhoods are consistent with Schelling dynamics: each household's mild preference for "some" same-type neighbors, iterated over decades of moves, produces stark global segregation.

From Grids to Networks

In a network setting, Schelling's model becomes a **rewiring** process:

- "neighbors" are graph-adjacent nodes
- "moving" becomes dropping a tie to a dissimilar contact and forming a new tie to a similar contact
- the threshold τ controls how much same-type concentration triggers rewiring

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- the threshold τ controls how much same-type concentration triggers rewiring

Implication for recommendation algorithms. If a platform suggests contacts who are *similar* to your current contacts, it acts as an automated Schelling rewirer — accelerating homophily-driven segregation even beyond what individual preferences would produce alone.

Quick Check

Translate Schelling's spatial model into a network setting.

1. What plays the role of "neighbor" in a network rather than a grid?
2. What plays the role of "moving"?
3. A social media platform recommends friends similar to your existing friends. In Schelling terms, what is the platform doing — and would you expect it to amplify or weaken homophily?

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1. Network neighbor: a graph-adjacent node — your immediate contacts.

2. Moving: rewiring — dropping ties to dissimilar contacts and forming new ones with similar contacts.

3. The platform is an automated Schelling rewirer. It surfaces similar people — equivalent to offering each agent a move toward a higher same-type fraction. This *amplifies* homophily: the local rule ("recommend similar") is mild, but at machine speed it produces sharp global outcomes (filter bubbles, echo chambers).

Takeaway

Global polarization can emerge without anyone explicitly aiming for it. Mild local preferences plus iterated local updates produce sharp large-scale structure — and recommendation algorithms now run this process at scale.

Wrap-Up — Lecture 05

L05 added **context and time** to the graph. Nodes now have attributes, ties have timestamps, and social settings become explicit foci.

Question	L05 answer
Is similarity more than chance?	Compare observed homophily to a population baseline.
Why are similar people connected?	Distinguish selection, socialization, and shared context.
Where do ties form?	Model foci with affiliation networks and co-occurrence graphs.
How do local rules scale up?	Use closure curves and Schelling dynamics to connect micro behavior to macro patterns.

The central lesson: a graph snapshot can show a pattern, but explaining the mechanism requires baselines, temporal data, context, or a generative model.

Reading

Chapter 4 of Easley & Kleinberg (2010) covers homophily, selection, and socialization. McPherson, Smith-Lovin & Cook (2001), *Birds of a Feather*, is the definitive review of homophily research. Christakis & Fowler (2007) is the Framingham obesity study; Lyons (2011) provides the methodological critique.

Image Credits & References

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